



# Databricks Data Engineer Professional Practice Exam

105 Questions • University • Computers

## QUIZZIZZ Worksheets

### Databricks Data Engineer Professional Practice Exam

Total questions: 105

Worksheet time: 53mins

Instructor name: Vo Hong

Name

Class

Date

1. An upstream system has been configured to pass the date for a given batch of data to the Databricks Jobs API as a parameter. The notebook to be scheduled will use this parameter to load data with the following code: `df = spark.read.format("parquet").load(f"/mnt/source/{(date)})`

Which code block should be used to create the date Python variable used in the above code block?

  - a) `input_dict = input()`  
`date= input_dict["date"]`
  - b) `import sys`  
`date = sys.argv[1]`
  - c) `date = dbutils.notebooks.getParam("date")`
  - d) `dbutils.widgets.text("date", "null")`  
`date = dbutils.widgets.get("date")`
  - e) `date = spark.conf.get("date")`
2. The Databricks workspace administrator has configured interactive clusters for each of the data engineering groups. To control costs, clusters are set to terminate after 30 minutes of inactivity. Each user should be able to execute workloads against their assigned clusters at any time of the day.

Assuming users have been added to a workspace but not granted any permissions, which of the following describes the minimal permissions a user would need to start and attach to an already configured cluster.

  - a) Cluster creation allowed, "Can Restart" privileges on the required cluster
  - b) Workspace Admin privileges, cluster creation allowed, "Can Attach To" privileges on the required cluster
  - c) "Can Restart" privileges on the required cluster
  - d) "Can Manage" privileges on the required cluster
  - e) Cluster creation allowed, "Can Attach To" privileges on the required cluster

3. When scheduling Structured Streaming jobs for production, which configuration automatically recovers from query failures and keeps costs low?

- |                                                                                           |                                                                                                |
|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| a) Cluster: New Job Cluster;<br>Retries: None;<br>Maximum Concurrent Runs: 1              | b) Cluster: Existing All-Purpose Cluster;<br>Retries: None;<br>Maximum Concurrent Runs: 1      |
| c) Cluster: New Job Cluster;<br>Retries: Unlimited;<br>Maximum Concurrent Runs: Unlimited | d) Cluster: Existing All-Purpose Cluster;<br>Retries: Unlimited;<br>Maximum Concurrent Runs: 1 |
| e) Cluster: New Job Cluster;<br>Retries: Unlimited;<br>Maximum Concurrent Runs: 1         |                                                                                                |

4. `SELECT MEAN(temperature), MAX(temperature), MIN(temperature)  
FROM recent_sensor_recordings  
GROUP BY sensor_id`

The data engineering team has configured a Databricks SQL query and alert to monitor the values in a Delta Lake table. The `recent_sensor_recordings` table contains an identifying `sensor_id` alongside the timestamp and temperature for the most recent 5 minutes of recordings.

The query on the left is used to create the alert:

The query is set to refresh each minute and always completes in less than 10 seconds. The alert is set to trigger when `mean (temperature) > 120`. Notifications are triggered to be sent at most every 1 minute.

If this alert raises notifications for 3 consecutive minutes and then stops, which statement must be true?

- |                                                                                                                        |                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| a) The maximum temperature recording for at least one sensor exceeded 120 on three consecutive executions of the query | b) The total average temperature across all sensors exceeded 120 on three consecutive executions of the query           |
| c) The <code>recent_sensor_recordings</code> table was unresponsive for three consecutive runs of the query            | d) The average temperature recordings for at least one sensor exceeded 120 on three consecutive executions of the query |
| e) The source query failed to update properly for three consecutive minutes and then restarted                         |                                                                                                                         |

5. A junior developer complains that the code in their notebook isn't producing the correct results in the development environment. A shared screenshot reveals that while they're using a notebook versioned with Databricks Repos, they're using a personal branch that contains old logic. The desired branch named dev-2.3.9 is not available from the branch selection dropdown.

Which approach will allow this developer to review the current logic for this notebook?

- a) Use Repos to merge the current branch and the dev-2.3.9 branch, then make a pull request to sync with the remote repository
- b) Use Repos to make a pull request use the Databricks REST API to update the current branch to dev-2.3.9
- c) Use Repos to checkout the dev-2.3.9 branch and auto-resolve conflicts with the current branch
- d) Use Repos to pull changes from the remote Git repository and select the dev-2.3.9 branch.
- e) Merge all changes back to the main branch in the remote Git repository and clone the repo again

6.

```
password = dbrutils.secrets.get(scope="db_creds", key="jdbc_password")
print(password)

df = (spark
      .read
      .format("jdbc")
      .option("url", connection)
      .option("dbtable", tablename)
      .option("user", username)
      .option("password", password)
      )
```

The security team is exploring whether or not the Databricks secrets module can be leveraged for connecting to an external database.

After testing the code with all Python variables being defined with strings, they upload the password to the secrets module and configure the correct permissions for the currently active user. They then modify their code to the following (leaving all other variables unchanged). (See screenshot on the left)

Which statement describes what will happen when the above code is executed?

- a) The connection to the external table will succeed; the string "REDACTED" will be printed.
- b) An interactive input box will appear in the notebook; if the right password is provided, the connection will succeed and the password will be printed in plain text.
- c) An interactive input box will appear in the notebook; if the right password is provided, the connection will succeed and the encoded password will be saved to DBFS.
- d) The connection to the external table will fail; the string "REDACTED" will be printed.
- e) The connection to the external table will succeed; the string value of password will be printed in plain text.

7.

```

from pyspark.sql.functions import current_date

model = mlflow.pyfunc.spark_udf(spark, model_uri="models:/churn/prod")
df = spark.table("customers")
columns = ["account_age", "time_since_last_seen", "app_rating"]
preds = (df.select(
    "customer_id",
    model(*columns).alias("predictions"),
    current_date().alias("date")
))

```

The data science team has created and logged a production model using MLflow. The following code c and applies the production model to output the predictions as a new DataFrame named preds with th "customer\_id LONG, predictions DOUBLE, date DATE".

The data science team would like predictions saved to a Delta Lake table with the ability to compare al across time. Churn predictions will be made at most once per day.

Which code block accomplishes this task while minimizing potential compute costs?

a) preds.write.mode("append").saveAsTable("churn\_preds") b) preds.write.format("delta").save("/pre

c)

```

(preds.writeStream
  .outputMode("overwrite")
  .option("checkpointPath", "/_checkpoints/churn_preds")
  .start("/preds/churn_preds")
)

```

d)

```

(preds.writeStream
  .outputMode("append")
  .option("checkpointPath", "/_checkpoints/churn_preds")
  .table("churn_preds")
)

```

e)

```

(preds.write
  .format("delta")
  .mode("overwrite")
  .saveAsTable("churn_preds")
)

```

```
8. (spark.read
    .format("parquet")
    .load(f"/mnt/raw_orders/{date}")
    .dropDuplicates(["customer_id", "order_id"])
    .write
    .mode("append")
    .saveAsTable("orders")
)
```

An upstream source writes Parquet data as hourly batches to directories named with the current date. A nightly batch job runs the following code to ingest all data from the previous day as indicated by the date variable:

Assume that the fields `customer_id` and `order_id` serve as a composite key to uniquely identify each order.

If the upstream system is known to occasionally produce duplicate entries for a single order hours apart, which statement is correct?

- a) Each write to the orders table will only contain unique records, and only those records without duplicates in the target table will be written.
- b) Each write to the orders table will only contain unique records, but newly written records may have duplicates already present in the target table.
- c) Each write to the orders table will run deduplication over the union of new and existing records, ensuring no duplicate records are present.
- d) Each write to the orders table will only contain unique records; if existing records with the same key are present in the target table, the operation will fail.
- e) Each write to the orders table will only contain unique records; if existing records with the same key are present in the target table, these records will be overwritten.

9.

```

Cmd 1
%python
countries_af = [x[0] for x in
spark.table("geo_lookup").filter("continent='AF'").select("country").collect()]

Cmd 2
%sql
CREATE VIEW sales_af AS
SELECT *
FROM sales
WHERE city IN countries_af
AND CONTINENT = "AF"

```

A junior member of the data engineering team is exploring the language interoperability of Databricks notebooks. The intended outcome of the below code is to register a view of all sales that occurred in countries on the continent of Africa that appear in the geo\_lookup table. Before executing the code, running SHOW TABLES on the current database indicates the database contains only two tables: geo\_lookup and sales.

Which statement correctly describes the outcome of executing these command cells in order in an interactive notebook?

- a) Cmd 1 will succeed and Cmd 2 will fail. countries\_af will be a Python variable containing a list of strings.
- b) Both commands will succeed. Executing show tables will show that countries\_af and sales\_af have been registered as views.
- c) Cmd 1 will succeed. Cmd 2 will search all accessible databases for a table or view named countries\_af: if this entity exists, Cmd 2 will succeed.
- d) Cmd 1 will succeed and Cmd 2 will fail. countries\_af will be a Python variable representing a PySpark DataFrame.
- e) Both commands will fail. No new variables, tables, or views will be created.

10. A Delta table of weather records is partitioned by date and has the below schema: date DATE, device\_id INT, temp FLOAT, latitude FLOAT, longitude FLOAT

To find all the records from within the Arctic Circle, you execute a query with the below filter: latitude > 66.3

Which statement describes how the Delta engine identifies which files to load?

- a) The Parquet file footers are scanned for min and max statistics for the latitude column
- b) All records are cached to attached storage and then the filter is applied
- c) All records are cached to an operational database and then the filter is applied
- d) The Hive metastore is scanned for min and max statistics for the latitude column
- e) The Delta log is scanned for min and max statistics for the latitude column

11. The data engineering team has configured a job to process customer requests to be forgotten (have their data deleted). All user data that needs to be deleted is stored in Delta Lake tables using default table settings.

The team has decided to process all deletions from the previous week as a batch job at 1am each Sunday. The total duration of this job is less than one hour. Every Monday at 3am, a batch job executes a series of VACUUM commands on all Delta Lake tables throughout the organization.

The compliance officer has recently learned about Delta Lake's time travel functionality. They are concerned that this might allow continued access to deleted data.

Assuming all delete logic is correctly implemented, which statement correctly addresses this concern?

- a) Because the default data retention threshold is 24 hours, data files containing deleted records will be retained until the VACUUM job is run the following day.
- b) Because Delta Lake time travel provides full access to the entire history of a table, deleted records can always be recreated by users with full admin privileges.
- c) Because Delta Lake's delete statements have ACID guarantees, deleted records will be permanently purged from all storage systems as soon as a delete job completes.
- d) Because the default data retention threshold is 7 days, data files containing deleted records will be retained until the VACUUM job is run 8 days later.
- e) Because the VACUUM command permanently deletes all files containing deleted records, deleted records may be accessible with time travel for around 24 hours.

12. 

```
{
  "name": "Ingest new data",
  "existing_cluster_id": "6015-954420-peace720",
  "notebook_task": {
    "notebook_path": "/Prod/ingest.py"
  }
}
```

A junior data engineer has configured a workload that posts the following JSON to the Databricks REST API endpoint 2.0/jobs/create. (SEE IMAGE ON THE LEFT)

Assuming that all configurations and referenced resources are available, which statement describes the result of executing this workload three times?

- a) The logic defined in the referenced notebook will be executed three times on the referenced existing all purpose cluster.
- b) Three new jobs named "Ingest new data" will be defined in the workspace, and they will each run once daily.
- c) The logic defined in the referenced notebook will be executed three times on new clusters with the configurations of the provided cluster ID.
- d) Three new jobs named "Ingest new data" will be defined in the workspace, but no jobs will be executed.
- e) One new job named "Ingest new data" will be defined in the workspace, but it will not be executed.



13. An upstream system is emitting change data capture (CDC) logs that are being written to a cloud object storage directory. Each record in the log indicates the change type (insert, update, or delete) and the values for each field after the change. The source table has a primary key identified by the field `pk_id`.

For auditing purposes, the data governance team wishes to maintain a full record of all values that have ever been valid in the source system. For analytical purposes, only the most recent value for each record needs to be recorded. The Databricks job to ingest these records occurs once per hour, but each individual record may have changed multiple times over the course of an hour.

Which solution meets these requirements?

- a) Ingest all log information into a bronze table; use `MERGE INTO` to insert, update, or delete the most recent entry for each `pk_id` into a silver table to recreate the current table state.
- b) Create a separate history table for each `pk_id` resolve the current state of the table by running a union all filtering the history tables for the most recent state.
- c) Iterate through an ordered set of changes to the table, applying each in turn; rely on Delta Lake's versioning ability to create an audit log.
- d) Use Delta Lake's change data feed to automatically process CDC data from an external system, propagating all changes to all dependent tables in the Lakehouse.
- e) Use `MERGE INTO` to insert, update, or delete the most recent entry for each `pk_id` into a bronze table, then propagate all changes throughout the system.

14. An hourly batch job is configured to ingest data files from a cloud object storage container where each batch represent all records produced by the source system in a given hour. The batch job to process these records into the Lakehouse is sufficiently delayed to ensure no late-arriving data is missed. The user\_id field represents a unique key for the data, which has the following schema: user\_id BIGINT, username STRING, user\_utc STRING, user\_region STRING, last\_login BIGINT, auto\_pay BOOLEAN, last\_updated BIGINT
- New records are all ingested into a table named account\_history which maintains a full record of all data in the same schema as the source. The next table in the system is named account\_current and is implemented as a Type 1 table representing the most recent value for each unique user\_id.
- Assuming there are millions of user accounts and tens of thousands of records processed hourly, which implementation can be used to efficiently update the described account\_current table as part of each hourly batch job?

- a) Overwrite the account\_current table with each batch using the results of a query against the account\_history table grouping by user\_id and filtering for the max value of last\_updated.
- b) Use Delta Lake version history to get the difference between the latest version of account\_history and one version prior, then write these records to account\_current.
- c) Use Auto Loader to subscribe to new files in the account\_history directory; configure a Structured Streaming trigger once job to batch update newly detected files into the account\_current table.
- d) Filter records in account\_history using the last\_updated field and the most recent hour processed, making sure to deduplicate on username; write a merge statement to update or insert the most recent value for each username.
- e) Filter records in account\_history using the last\_updated field and the most recent hour processed, as well as the max last\_login by user\_id write a merge statement to update or insert the most recent value for each user\_id.

15. A Delta Lake table in the Lakehouse named `customer_churn_params` is used in churn prediction by the machine learning team. The table contains information about customers derived from a number of upstream sources. Currently, the data engineering team populates this table nightly by overwriting the table with the current valid values derived from upstream data sources.

The churn prediction model used by the ML team is fairly stable in production. The team is only interested in making predictions on records that have changed in the past 24 hours.

Which approach would simplify the identification of these changed records?

- a) Replace the current override logic with a merge statement to modify only those records that have changed; write logic to make predictions on the changed records identified by the change data feed.
- b) Apply the churn model to all rows in the `customer_churn_params` table, but implement logic to perform an upsert into the predictions table that ignores rows where predictions have not changed.
- c) Convert the batch job to a Structured Streaming job using the complete output mode; configure a Structured Streaming job to read from the `customer_churns_params` table and incrementally predict against the churn model.
- d) Modify the overwrite logic to include a field populated by calling `spark.sql.functions.current_timestamp()` as data are being written; use this fields to identify records written on a particular date.

16. 

```
CREATE TABLE recent_orders AS (  
  SELECT a.user_id, a.email, b.order_id, b.order_date  
  FROM  
    (SELECT user_id, email  
     FROM users) a  
    INNER JOIN  
    (SELECT user_id, order_id, order_date  
     FROM orders  
     WHERE order_date >= (current_date() - 7)) b  
    ON a.user_id = b.user_id  
)
```

A table is registered with the following code: (SEE IMAGE ON THE LEFT)

Both users and orders are Delta Lake tables. Which statement describes the results of querying recent\_orders?

- a) Results will be computed and cached when the table is defined; these cached results will incrementally update as new records are inserted into source tables.
- b) All logic will execute at query time and return the result of joining the valid versions of the source tables at the time the query finishes.
- c) The versions of each source table will be stored in the table transaction log; query results will be saved to DBFS with each query.
- d) All logic will execute when the table is defined and store the result of joining tables to the DBFS; this stored data will be returned when the table is queried.
- e) All logic will execute at query time and return the result of joining the valid versions of the source tables at the time the query began.

17. A production workload incrementally applies updates from an external Change Data Capture feed to a Delta Lake table as an always-on Structured Stream job. When data was initially migrated for this table, OPTIMIZE was executed and most data files were resized to 1 GB. Auto Optimize and Auto Compaction were both turned on for the streaming production job. Recent review of data files shows that most data files are under 64 MB, although each partition in the table contains at least 1 GB of data and the total table size is over 10 TB.

Which of the following likely explains these smaller file sizes?

- a) Databricks has autotuned to a smaller target file size based on the overall size of data in the table
- b) Z-order indices calculated on the table are preventing file compaction
- c) Databricks has autotuned to a smaller target file size to reduce duration of MERGE operations
- d) Bloom filter indices calculated on the table are preventing file compaction
- e) Databricks has autotuned to a smaller target file size based on the amount of data in each partition

18. Which statement regarding stream-static joins and static Delta tables is correct?
- a) Each microbatch of a stream-static join will use the most recent version of the static Delta table as of the job's initialization.
  - b) Stream-static joins cannot use static Delta tables because of consistency issues.
  - c) Each microbatch of a stream-static join will use the most recent version of the static Delta table as of each microbatch.
  - d) The checkpoint directory will be used to track updates to the static Delta table.
  - e) The checkpoint directory will be used to track state information for the unique keys present in the join.

19. 

```
df.withWatermark("event_time", "10 minutes")  
  .groupBy(  
    _____,  
    "device_id"  
  )  
  .agg(  
    avg("temp").alias("avg_temp"),  
    avg("humidity").alias("avg_humidity")  
  )  
  .writeStream  
  .format("delta")  
  .saveAsTable("sensor_avg")
```

A junior data engineer has been asked to develop a streaming data pipeline with a grouped aggregation using DataFrame df. The pipeline needs to calculate the average humidity and average temperature for each non-overlapping five-minute interval. Events are recorded once per minute per device.

Streaming DataFrame df has the following schema:

"device\_id INT, event\_time TIMESTAMP, temp FLOAT, humidity FLOAT"

Code block: (SEE IMAGE ON THE LEFT)

Choose the response that correctly fills in the blank within the code block to complete this task.

- a) window("event\_time", "5 minutes").alias("time")
- b) lag("event\_time", "10 minutes").alias("time")
- c) to\_interval("event\_time", "5 minutes").alias("time")
- d) "event\_time"
- e) window("event\_time", "10 minutes").alias("time")

20.

```
./bronze
├── __checkpoint
├── __delta_log
├── year_week=2020_01
├── year_week=2020_02
└── ...
```

A data architect has designed a system in which two Structured Streaming jobs will concurrently write to a single bronze Delta table. Each job is subscribing to a different topic from an Apache Kafka source, but they will write data with the same schema. To keep the directory structure simple, a data engineer has decided to nest a checkpoint directory to be shared by both streams.

The proposed directory structure is displayed below: (SEE IMAGE ON THE LEFT)

Which statement describes whether this checkpoint directory structure is valid for the given scenario and why?

- a) No; each of the streams needs to have its own checkpoint directory.
- b) Yes; both of the streams can share a single checkpoint directory.
- c) Yes; Delta Lake supports infinite concurrent writers.
- d) No; only one stream can write to a Delta Lake table.
- e) No; Delta Lake manages streaming checkpoints in the transaction log.

21. A Structured Streaming job deployed to production has been experiencing delays during peak hours of the day. At present, during normal execution, each microbatch of data is processed in less than 3 seconds. During peak hours of the day, execution time for each microbatch becomes very inconsistent, sometimes exceeding 30 seconds. The streaming write is currently configured with a trigger interval of 10 seconds.

Holding all other variables constant and assuming records need to be processed in less than 10 seconds, which adjustment will meet the requirement?

- a) Decrease the trigger interval to 5 seconds; triggering batches more frequently allows idle executors to begin processing the next batch while longer running tasks from previous batches finish.
- b) The trigger interval cannot be modified without modifying the checkpoint directory; to maintain the current stream state, increase the number of shuffle partitions to maximize parallelism.
- c) Decrease the trigger interval to 5 seconds; triggering batches more frequently may prevent records from backing up and large batches from causing spill.
- d) Increase the trigger interval to 30 seconds; setting the trigger interval near the maximum execution time observed for each batch is always best practice to ensure no records are dropped.
- e) Use the trigger once option and configure a Databricks job to execute the query every 10 seconds; this ensures all backlogged records are processed with each batch.

22. Which statement describes Delta Lake Auto Compaction?

- a) An asynchronous job runs after the write completes to detect if files could be further compacted; if yes, an OPTIMIZE job is executed toward a default of 128 MB.
- b) Optimized writes use logical partitions instead of directory partitions; because partition boundaries are only represented in metadata, fewer small files are written.
- c) Data is queued in a messaging bus instead of committing data directly to memory; all data is committed from the messaging bus in one batch once the job is complete.
- d) An asynchronous job runs after the write completes to detect if files could be further compacted; if yes, an OPTIMIZE job is executed toward a default of 1 GB.
- e) Before a Jobs cluster terminates, OPTIMIZE is executed on all tables modified during the most recent job.

23. Which statement characterizes the general programming model used by Spark Structured Streaming?
- a) Structured Streaming models new data arriving in a data stream as new rows appended to an unbounded table.
  - b) Structured Streaming relies on a distributed network of nodes that hold incremental state values for cached stages.
  - c) Structured Streaming is implemented as a messaging bus and is derived from Apache Kafka.
  - d) Structured Streaming leverages the parallel processing of GPUs to achieve highly parallel data throughput.
  - e) Structured Streaming uses specialized hardware and I/O streams to achieve sub-second latency for data transfer.
24. Which configuration parameter directly affects the size of a spark-partition upon ingestion of data into Spark?
- a) `spark.sql.files.maxPartitionBytes`
  - b) `spark.sql.adaptive.coalescePartitions.minPartitionNum`
  - c) `spark.sql.autoBroadcastJoinThreshold`
  - d) `spark.sql.adaptive.advisoryPartitionSizeInBytes`
  - e) `spark.sql.files.openCostInBytes`
25. A Spark job is taking longer than expected. Using the Spark UI, a data engineer notes that the Min, Median, and Max Durations for tasks in a particular stage show the minimum and median time to complete a task as roughly the same, but the max duration for a task to be roughly 100 times as long as the minimum. Which situation is causing increased duration of the overall job?
- a) Credential validation errors while pulling data from an external system.
  - b) Task queueing resulting from improper thread pool assignment.
  - c) Skew caused by more data being assigned to a subset of spark-partitions.
  - d) Network latency due to some cluster nodes being in different regions from the source data
  - e) Spill resulting from attached volume storage being too small.



26. Each configuration below is identical to the extent that each cluster has 400 GB total of RAM, 160 total cores and only one Executor per VM.

Given a job with at least one wide transformation, which of the following cluster configurations will result in maximum performance?

- |                        |                       |
|------------------------|-----------------------|
| a) • Total VMs: 1      | b) • Total VMs: 4     |
| • 400 GB per Executor  | • 100 GB per Executor |
| • 160 Cores / Executor | • 40 Cores/Executor   |
| c) • Total VMs: 8      | d) • Total VMs: 2     |
| • 50 GB per Executor   | • 200 GB per Executor |
| • 20 Cores / Executor  | • 80 Cores / Executor |
| e) • Total VMs: 16     |                       |
| • 25 GB per Executor   |                       |
| • 10 Cores/Executor    |                       |

27. 

```
MERGE INTO events
USING new_events
ON events.event_id = new_events.event_id
WHEN NOT MATCHED
  INSERT *
```

A junior data engineer on your team has implemented the following code block.

(SEE IMAGE ON THE LEFT)

The view new\_events contains a batch of records with the same schema as the events Delta table. The event\_id field serves as a unique key for this table.

When this query is executed, what will happen with new records that have the same event\_id as an existing record.

- |                      |                       |
|----------------------|-----------------------|
| a) They are updated. | b) They are deleted.  |
| c) They are merged.  | d) They are inserted. |
| e) They are ignored. |                       |

28.

```
from pyspark.sql.functions import col

(spark.read.format("delta")
 .option("readChangeFeed", "true")
 .option("startingVersion", 0)
 .table("bronze")
 .filter(col("_change_type").isin(["update_postimage", "insert"]))
 .write
 .mode("append")
 .table("bronze_history_type1")
)
```

A junior data engineer seeks to leverage Delta Lake's Change Data Feed functionality to create a Type 1 table representing all of the values that have ever been valid for all rows in a bronze table created with the property `delta.enableChangeDataFeed = true`. They plan to execute the following code as a daily job: (SEE IMAGE ON THE LEFT)

Which statement describes the execution and results of running the above query multiple times?

- a) Each time the job is executed, the entire available history of inserted or updated records will be appended to the target table, resulting in many duplicate entries.
- b) Each time the job is executed, newly updated records will be merged into the target table, overwriting previous values with the same primary keys.
- c) Each time the job is executed, the differences between the original and current versions are calculated; this may result in duplicate entries for some records.
- d) Each time the job is executed, only those records that have been inserted or updated since the last execution will be appended to the target table, giving the desired result.
- e) Each time the job is executed, the target table will be overwritten using the entire history of inserted or updated records, giving the desired result.

29. A new data engineer notices that a critical field was omitted from an application that writes its Kafka source to Delta Lake. This happened even though the critical field was in the Kafka source. That field was further missing from data written to dependent, long-term storage. The retention threshold on the Kafka service is seven days. The pipeline has been in production for three months.

Which describes how Delta Lake can help to avoid data loss of this nature in the future?

- a) Delta Lake automatically checks that all fields present in the source data are included in the ingestion layer.
  - b) Delta Lake schema evolution can retroactively calculate the correct value for newly added fields, as long as the data was in the original source.
  - c) The Delta log and Structured Streaming checkpoints record the full history of the Kafka producer.
  - d) Ingesting all raw data and metadata from Kafka to a bronze Delta table creates a permanent, replayable history of the data state.
  - e) Data can never be permanently dropped or deleted from Delta Lake, so data loss is not possible under any circumstance.
30. A nightly job ingests data into a Delta Lake table using the following code: (SEE IMAGE ON THE LEFT)

The next step in the pipeline requires a function that returns an object that can be used to manipulate new records that have not yet been processed to the next table in the pipeline. Which code snippet completes this function definition?

def new\_records():

a) 

```
return spark.read
    .table("bronze")
    .filter(col("source_file") == f"/mnt/daily_batch/{year}/{month}/{day}")
    
```

b) 

```
return (spark.read
    .table("bronze")
    .filter(col("ingest_time") == current_timestamp())
    )
    
```

c) `return spark.readStream.table("bronze")`

d) `return spark.read.option("readChangeFeed", "true").table ("bronze")`

e) `return spark.readStream.load("bronze")`

31. A junior data engineer is working to implement logic for a Lakehouse table named `silver_device_recordings`. The source data contains 100 unique fields in a highly nested JSON structure.

The `silver_device_recordings` table will be used downstream to power several production monitoring dashboards and a production model. At present, 45 of the 100 fields are being used in at least one of these applications.

The data engineer is trying to determine the best approach for dealing with schema declaration given the highly-nested structure of the data and the numerous fields.

Which of the following accurately presents information about Delta Lake and Databricks that may impact their decision-making process?

- a) Because Delta Lake uses Parquet for data storage, data types can be easily evolved by just modifying file footer information in place.
- b) Human labor in writing code is the largest cost associated with data engineering workloads; as such, automating table declaration logic should be a priority in all migration workloads.
- c) Because Databricks will infer schema using types that allow all observed data to be processed, setting types manually provides greater assurance of data quality enforcement.
- d) Schema inference and evolution on Databricks ensure that inferred types will always accurately match the data types used by downstream systems.
- e) The Tungsten encoding used by Databricks is optimized for storing string data; newly-added native support for querying JSON strings means that string types are always most efficient.

32.

```

accountDF = spark.table("accounts")
orderDF = spark.table("orders")
itemDF = spark.table("items")

orderWithItemDF = (orderDF.join(
    itemDF,
    orderDF.itemID == itemDF.itemID)
    .select(
        orderDF.accountID,
        orderDF.itemID,
        itemDF.itemName))

finalDF = (accountDF.join(
    orderWithItemDF,
    accountDF.accountID == orderWithItemDF.accountID)
    .select(
        orderWithItemDF["*"],
        accountDF.city))

(finalDF.write
    .mode("overwrite")
    .table("enriched_itemized_orders_by_account"))

```

The data engineering team maintains the following code: (SEE IMAGE ON THE LEFT)

Assuming that this code produces logically correct results and the data in the source tables has been de-duplicated and validated, which statement describes what will occur when this code is executed?

- a) A batch job will update the enriched\_itemized\_orders\_by\_account table, replacing only those rows that have different values than the current version of the table, using accountID as the primary key.
- b) An incremental job will leverage information in the state store to identify unjoined rows in the source tables and write these rows to the enriched\_itemized\_orders\_by\_account table.
- c) An incremental job will detect if new rows have been written to any of the source tables; if new rows are detected, all results will be recalculated and used to overwrite the enriched\_itemized\_orders\_by\_account table.
- d) No computation will occur until enriched\_itemized\_orders\_by\_account is queried; upon query materialization, results will be calculated using the current valid version of data in each of the three tables referenced in the join logic.
- e) The enriched\_itemized\_orders\_by\_account table will be overwritten using the current valid version of data in each of the three tables referenced in the join logic.

33. The data engineering team is migrating an enterprise system with thousands of tables and views into the Lakehouse. They plan to implement the target architecture using a series of bronze, silver, and gold tables. Bronze tables will almost exclusively be used by production data engineering workloads, while silver tables will be used to support both data engineering and machine learning workloads. Gold tables will largely serve business intelligence and reporting purposes. While personal identifying information (PII) exists in all tiers of data, pseudonymization and anonymization rules are in place for all data at the silver and gold levels.

The organization is interested in reducing security concerns while maximizing the ability to collaborate across diverse teams.

Which statement exemplifies best practices for implementing this system?

- a) Storing all production tables in a single database provides a unified view of all data assets available throughout the Lakehouse, simplifying discoverability by granting all users view privileges on this database.
- b) Working in the default Databricks database provides the greatest security when working with managed tables, as these will be created in the DBFS root.
- c) Because databases on Databricks are merely a logical construct, choices around database organization do not impact security or discoverability in the Lakehouse.
- d) Because all tables must live in the same storage containers used for the database they're created in, organizations should be prepared to create between dozens and thousands of databases depending on their data isolation requirements.
- e) Isolating tables in separate databases based on data quality tiers allows for easy permissions management through database ACLs and allows physical separation of default storage locations for managed tables.

34. The data architect has mandated that all tables in the Lakehouse should be configured as external Delta Lake tables.

Which approach will ensure that this requirement is met?

- a) When configuring an external data warehouse for all table storage, leverage Databricks for all ELT.
- b) Whenever a table is being created, make sure that the LOCATION keyword is used.
- c) When the workspace is being configured, make sure that external cloud object storage has been mounted.
- d) Whenever a database is being created, make sure that the LOCATION keyword is used
- e) When tables are created, make sure that the EXTERNAL keyword is used in the CREATE TABLE statement.

35. To reduce storage and compute costs, the data engineering team has been tasked with curating a series of aggregate tables leveraged by business intelligence dashboards, customer-facing applications, production machine learning models, and ad hoc analytical queries.

The data engineering team has been made aware of new requirements from a customer-facing application, which is the only downstream workload they manage entirely. As a result, an aggregate table used by numerous teams across the organization will need to have a number of fields renamed, and additional fields will also be added.

Which of the solutions addresses the situation while minimally interrupting other teams in the organization without increasing the number of tables that need to be managed?

- a) Send all users notice that the schema for the table will be changing; include in the communication the logic necessary to revert the new table schema to match historic queries.
- b) Configure a new table with all the requisite fields and new names and use this as the source for the customer-facing application; create a view that maintains the original data schema and table name by aliasing select fields from the new table.
- c) Replace the current table definition with a logical view defined with the query logic currently writing the aggregate table; create a new table to power the customer-facing application.
- d) Create a new table with the required schema and new fields and use Delta Lake's deep clone functionality to sync up changes committed to one table to the corresponding table.
- e) Add a table comment warning all users that the table schema and field names will be changing on a given date; overwrite the table in place to the specifications of the customer-facing application.

36. A Delta Lake table representing metadata about content posts from users has the following schema: user\_id LONG, post\_text STRING, post\_id STRING, longitude FLOAT, latitude FLOAT, post\_time TIMESTAMP, date DATE
- This table is partitioned by the date column. A query is run with the following filter: longitude < 20 & longitude > -20
- Which statement describes how data will be filtered?
- a) No file skipping will occur because the optimizer does not know the relationship between the partition column and the longitude.
  - b) Statistics in the Delta Log will be used to identify partitions that might include files in the filtered range.
  - c) The Delta Engine will use row-level statistics in the transaction log to identify the files that meet the filter criteria.
  - d) The Delta Engine will scan the parquet file footers to identify each row that meets the filter criteria.
  - e) Statistics in the Delta Log will be used to identify data files that might include records in the filtered range.
37. A small company based in the United States has recently contracted a consulting firm in India to implement several new data engineering pipelines to power artificial intelligence applications. All the company's data is stored in regional cloud storage in the United States. The workspace administrator at the company is uncertain about where the Databricks workspace used by the contractors should be deployed.
- Assuming that all data governance considerations are accounted for, which statement accurately informs this decision?
- a) Databricks leverages user workstations as the driver during interactive development; as such, users should always use a workspace deployed in a region they are physically near.
  - b) Databricks runs HDFS on cloud volume storage; as such, cloud virtual machines must be deployed in the region where the data is stored.
  - c) Cross-region reads and writes can incur significant costs and latency; whenever possible, compute should be deployed in the same region the data is stored.
  - d) Databricks workspaces do not rely on any regional infrastructure; as such, the decision should be made based upon what is most convenient for the workspace administrator.
  - e) Databricks notebooks send all executable code from the user's browser to virtual machines over the open internet; whenever possible, choosing a workspace region near the end users is the most secure.



38. 

```
ALTER TABLE activity_details
ADD CONSTRAINT valid_coordinates
CHECK (
    latitude >= -90 AND
    latitude <= 90 AND
    longitude >= -180 AND
    longitude <= 180);
```

The downstream consumers of a Delta Lake table have been complaining about data quality issues impacting performance in their applications. Specifically, they have complained that invalid latitude and longitude values in the activity\_details table have been breaking their ability to use other geolocation processes.

A junior engineer has written the following code to add CHECK constraints to the Delta Lake table: (SEE IMAGE ON THE LEFT)

A senior engineer has confirmed the above logic is correct and the valid ranges for latitude and longitude are provided, but the code fails when executed.

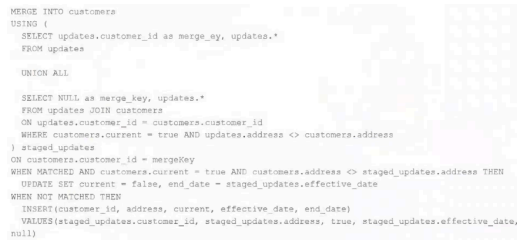
Which statement explains the cause of this failure?

- a) The current table schema does not contain the field valid\_coordinates; schema evolution will need to be enabled before altering the table to add a constraint.
- b) The activity\_details table already exists; CHECK constraints can only be added during initial table creation.
- c) The activity\_details table already contains records that violate the constraints; all existing data must pass CHECK constraints in order to add them to an existing table.
- d) Because another team uses this table to support a frequently running application, two-phase locking is preventing the operation from committing.
- e) The activity\_details table already contains records; CHECK constraints can only be added prior to inserting values into a table.

39. Which of the following is true of Delta Lake and the Lakehouse?

- a) Because Parquet compresses data row by row, strings will only be compressed when a character is repeated multiple times.
- b) Views in the Lakehouse maintain a valid cache of the most recent versions of source tables at all times.
- c) Delta Lake automatically collects statistics on the first 32 columns of each table which are leveraged in data skipping based on query filters.
- d) Z-order can only be applied to numeric values stored in Delta Lake tables.
- e) Primary and foreign key constraints can be leveraged to ensure duplicate values are never entered into a dimension table.

40.



```

MERGE INTO customers
USING (
  SELECT updates.customer_id as merge_key, updates.*
  FROM updates
) staged_updates
ON customers.customer_id = merge_key
WHEN MATCHED AND customers.current = true AND updates.address <> customers.address THEN
  UPDATE SET current = false, end_date = staged_updates.effective_date
WHEN NOT MATCHED THEN
  INSERT(customer_id, address, current, effective_date, end_date)
VALUES(staged_updates.customer_id, staged_updates.address, true, staged_updates.effective_date,
null)

```

The view updates represents an incremental batch of all newly ingested data to be inserted or updated in the customers table.

The following logic is used to process these records. (SEE IMAGE ON THE LEFT)

Which statement describes this implementation?

- a) The customers table is implemented as a Type 1 table; old values are overwritten by new values and no history is maintained.
  - b) The customers table is implemented as a Type 3 table; old values are maintained as a new column alongside the current value.
  - c) The customers table is implemented as a Type 2 table; old values are overwritten and new customers are appended.
  - d) The customers table is implemented as a Type 2 table; old values are maintained but marked as no longer current and new values are inserted.
  - e) The customers table is implemented as a Type 0 table; all writes are append only with no changes to existing values.
41. The DevOps team has configured a production workload as a collection of notebooks scheduled to run daily using the Jobs UI. A new data engineering hire is onboarding to the team and has requested access to one of these notebooks to review the production logic. What are the maximum notebook permissions that can be granted to the user without allowing accidental changes to production code or data?
- a) Can Run
  - b) No permissions
  - c) Can Manage
  - d) Can Read
  - e) Can Edit

42. 

```
CREATE VIEW email_ltv AS
SELECT
CASE WHEN
  is_member('marketing') THEN email
  ELSE 'REDACTED'
END AS email,
ltv
FROM user_ltv
```

A table named user\_ltv is being used to create a view that will be used by data analysts on various teams. Users in the workspace are configured into groups, which are used for setting up data access using ACLs.

The user\_ltv table has the following schema:

email STRING, age INT, ltv INT

The following view definition is executed: (SEE IMAGE ON THE LEFT)

An analyst who is not a member of the marketing group executes the following query:

SELECT \* FROM email\_ltv -

Which statement describes the results returned by this query?

- a) The email and ltv columns will be returned with the values in user\_ltv.
- b) Only the email and ltv columns will be returned; the email column will contain the string "REDACTED" in each row.
- c) Only the email and ltv columns will be returned; the email column will contain all null values.
- d) Three columns will be returned, but one column will be named "REDACTED" and contain only null values.
- e) The email.age, and ltv columns will be returned with the values in user\_ltv.

43. 

```
CREATE TABLE dev.pii_test
(id INT, name STRING COMMENT "PII")
COMMENT "Contains PII"
TBLPROPERTIES ('contains_pii' = True)
```

The data governance team has instituted a requirement that all tables containing Personal Identifiable Information (PII) must be clearly annotated. This includes adding column comments, table comments, and setting the custom table property "contains\_pii" = true.

The following SQL DDL statement is executed to create a new table: (SEE IMAGE ON THE LEFT)

Which command allows manual confirmation that these three requirements have been met?

- a) SHOW TBLPROPERTIES dev.pii\_test
- b) DESCRIBE HISTORY dev.pii\_test
- c) SHOW TABLES dev
- d) DESCRIBE EXTENDED dev.pii\_test
- e) DESCRIBE DETAIL dev.pii\_test

44.

Original query:

```
df.groupBy("item")
  .agg(count("item").alias("total_count"),
       mean("sale_price").alias("avg_price"))
  .writeStream
  .outputMode("complete")
  .option("checkpointLocation", "/item_agg/__checkpoint")
  .start("/item_agg")
```

Proposed query:

```
df.groupBy("item")
  .agg(count("item").alias("total_count"),
       mean("sale_price").alias("avg_price"))
  .writeStream
  .outputMode("complete")
  .option("checkpointLocation", "/item_agg/__checkpoint")
  .start("/item_agg")
```

A data team's Structured Streaming job is configured to calculate running aggregates for item sales to update a downstream marketing dashboard. The marketing team has introduced a new promotion, and they would like to add a new field to track the number of times this promotion code is used for each item. A junior data engineer suggests updating the existing query as follows. Note that proposed changes are in bold.

(See image in the left)

Which step must also be completed to put the proposed query into production?

- a) Increase the shuffle partitions to account for additional aggregates
- b) Run `REFRESH TABLE delta.`/item_agg``
- c) Register the data in the `"/item_agg"` directory to the Hive metastore
- d) Remove `.option('mergeSchema', 'true')` from the streaming write
- e) Specify a new `checkpointLocation`

45. The data engineering team is configuring environments for development, testing, and production before beginning migration on a new data pipeline. The team requires extensive testing on both the code and data resulting from code execution, and the team wants to develop and test against data as similar to production data as possible.

A junior data engineer suggests that production data can be mounted to the development and testing environments, allowing pre-production code to execute against production data. Because all users have admin privileges in the development environment, the junior data engineer has offered to configure permissions and mount this data for the team.

Which statement captures best practices for this situation?

- a) Because access to production data will always be verified using passthrough credentials, it is safe to mount data to any Databricks development environment.
- b) Because Delta Lake versions all data and supports time travel, it is not possible for user error or malicious actors to permanently delete production data; as such, it is generally safe to mount production data anywhere.
- c) All development, testing, and production code and data should exist in a single, unified workspace; creating separate environments for testing and development complicates administrative overhead.
- d) In environments where interactive code will be executed, production data should only be accessible with read permissions; creating isolated databases for each environment further reduces risks.
- e) As long as code in the development environment declares `USE dev_db` at the top of each notebook, there is no possibility of inadvertently committing changes back to production data sources.

46. A data engineer, User A, has promoted a pipeline to production by using the REST API to programmatically create several jobs. A DevOps engineer, User B, has configured an external orchestration tool to trigger job runs through the REST API. Both users authorized the REST API calls using their personal access tokens.
- A workspace admin, User C, inherits responsibility for managing this pipeline. User C uses the Databricks Jobs UI to take "Owner" privileges of each job. Jobs continue to be triggered using the credentials and tooling configured by User B.

An application has been configured to collect and parse run information returned by the REST API. Which statement describes the value returned in the `creator_user_name` field?

- a) Once User C takes "Owner" privileges, their email address will appear in this field; prior to this, User B's email address will appear in this field.
- b) Once User C takes "Owner" privileges, their email address will appear in this field; prior to this, User A's email address will appear in this field.
- c) User A's email address will always appear in this field, as they still own the underlying notebooks.
- d) User C will only ever appear in this field if they manually trigger the job, otherwise it will indicate User B.
- e) User B's email address will always appear in this field, as their credentials are always used to trigger the run.

47.

```
Cmd 1
rawDF = spark.table("raw_data")

Cmd 2
rawDF.printSchema()

Cmd 3
flattenedDF = rawDF.select("?", "values.*")

Cmd 4
finalDF = flattenedDF.drop("values")

Cmd 5
display(finalDF)

Cmd 6
finalDF.write.mode("append").saveAsTable("flat_data")
```

A member of the data engineering team has submitted a short notebook that they wish to schedule as part of a larger data pipeline. Assume that the commands provided below produce the logically correct results when run as presented.

(See image on the left)

Which command should be removed from the notebook before scheduling it as a job?

- a) Cmd6
- b) Cmd2
- c) Cmd5
- d) Cmd4
- e) Cmd3

48. Which statement regarding Spark configuration on the Databricks platform is true?

- a) Spark configuration properties set for an interactive cluster with the Clusters UI will impact all notebooks attached to that cluster.
- b) When the same Spark configuration property is set for an interactive cluster and a notebook attached to that cluster, the notebook setting will always be ignored.
- c) Spark configurations set within a notebook will affect all SparkSessions attached to the same interactive cluster.
- d) Spark configuration properties can only be set for an interactive cluster by creating a global init script.
- e) The Databricks REST API can be used to modify the Spark configuration properties for an interactive cluster without interrupting jobs currently running on the cluster.

49. The business reporting team requires that data for their dashboards be updated every hour. The total processing time for the pipeline that extracts transforms, and loads the data for their pipeline runs in 10 minutes. Assuming normal operating conditions, which configuration will meet their service-level agreement requirements with the lowest cost?
- a) Manually trigger a job anytime the business reporting team refreshes their dashboards
  - b) Schedule a Structured Streaming job with a trigger interval of 60 minutes
  - c) Schedule a job to execute the pipeline once an hour on a dedicated interactive cluster
  - d) Configure a job that executes every time new data lands in a given directory
  - e) Schedule a job to execute the pipeline once an hour on a new job cluster

50. A Databricks SQL dashboard has been configured to monitor the total number of records present in a collection of Delta Lake tables using the following query pattern:

```
SELECT COUNT (*) FROM table -
```

Which of the following describes how results are generated each time the dashboard is updated?

- a) The total count of rows is calculated by scanning all data files
- b) The total count of records is calculated from the parquet file metadata
- c) The total count of records is calculated from the Delta transaction logs
- d) The total count of rows will be returned from cached results unless REFRESH is run
- e) The total count of records is calculated from the Hive metastore



51. A Delta Lake table was created with the below query:

(See image on the left)

Consider the following query:

```
DROP TABLE prod.sales_by_store -
```

If this statement is executed by a workspace admin, which result will occur?

- a) Data will be marked as deleted but still recoverable with Time Travel.
- b) The table will be removed from the catalog but the data will remain in storage.
- c) The table will be removed from the catalog and the data will be deleted.
- d) Nothing will occur until a COMMIT command is executed.
- e) An error will occur because Delta Lake prevents the deletion of production data.

52. A developer has successfully configured their credentials for Databricks Repos and cloned a remote Git repository. They do not have privileges to make changes to the main branch, which is the only branch currently visible in their workspace.

Which approach allows this user to share their code updates without the risk of overwriting the work of their teammates?

- a) Use Repos to pull changes from the remote Git repository; commit and push changes to a branch that appeared as changes were pulled.
- b) Use Repos to checkout all changes and send the git diff log to the team.
- c) Use Repos to merge all differences and make a pull request back to the remote repository.
- d) Use Repos to create a fork of the remote repository, commit all changes, and make a pull request on the source repository.
- e) Use Repos to create a new branch, commit all changes, and push changes to the remote Git repository.

53.

```
password = dbutils.secrets.get(scope="db_creds", key="jdbc_password")  
print(password)  
  
df = (spark  
    .read  
    .format("jdbc")  
    .option("url", connection)  
    .option("dbtable", tablename)  
    .option("user", username)  
    .option("password", password)  
    )
```

he security team is exploring whether or not the Databricks secrets module can be leveraged for connecting to an external database.

After testing the code with all Python variables being defined with strings, they upload the password to the secrets module and configure the correct permissions for the currently active user. They then modify their code to the following (leaving all other variables unchanged).

(See image on the left)

Which statement describes what will happen when the above code is executed?

- |                                                                                                                                                                         |                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) The connection to the external table will succeed; the string value of password will be printed in plain text.                                                       | b) An interactive input box will appear in the notebook; if the right password is provided, the connection will succeed and the encoded password will be saved to DBFS. |
| c) An interactive input box will appear in the notebook; if the right password is provided, the connection will succeed and the password will be printed in plain text. | d) The connection to the external table will succeed; the string "REDACTED" will be printed.                                                                            |

54. 

```
model = mlflow.pyfunc.spark_udf(spark, model_uri="models:/churn/prod")
df = spark.table("customers")
columns = ["account_age", "time_since_last_seen", "app_rating"]
```

The data science team has created and logged a production model using MLflow. The model accepts a list of column names and returns a new column of type DOUBLE.

The following code correctly imports the production model, loads the customers table containing the customer\_id key column into a DataFrame, and defines the feature columns needed for the model.

(See image on the left)

Which code block will output a DataFrame with the schema "customer\_id LONG, predictions DOUBLE"?

- |                                                                                               |                                                                                                   |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| a) <code>df.select("customer_id",<br/>model(*columns).alias("predictions"))</code>            | b) <code>df.select("customer_id",<br/>pandas_udf(model,<br/>columns).alias("predictions"))</code> |
| c) <code>model.predict(df, columns)</code>                                                    | d) <code>df.apply(model,<br/>columns).select("customer_id, predictions")</code>                   |
| e) <code>df.map(lambda<br/>x:model(x[columns])).select("customer_id,<br/>predictions")</code> |                                                                                                   |

55.

```

Cmd 1
%python
countries_af = [x[0] for x in
spark.table("geo_lookup").filter("continent='AF'").select("country").collect()]

Cmd 2

%sql
CREATE VIEW sales_af AS
SELECT *
FROM sales
WHERE city IN countries_af
AND CONTINENT = "AF"

```

A junior member of the data engineering team is exploring the language interoperability of Databricks notebooks. The intended outcome of the below code is to register a view of all sales that occurred in countries on the continent of Africa that appear in the geo\_lookup table.

Before executing the code, running SHOW TABLES on the current database indicates the database contains only two tables: geo\_lookup and sales.

(See image on the left)

What will be the outcome of executing these command cells in order in an interactive notebook?

- |                                                                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| a) Cmd 1 will succeed and Cmd 2 will fail. countries_af will be a Python variable containing a list of strings.                                      | b) Cmd 1 will succeed and Cmd 2 will fail. countries_af will be a Python variable representing a PySpark DataFrame.          |
| c) Cmd 1 will succeed. Cmd 2 will search all accessible databases for a table or view named countries_af: if this entity exists, Cmd 2 will succeed. | d) Both commands will succeed. Executing SHOW TABLES will show that countries_af and sales_af have been registered as views. |

56. The data science team has requested assistance in accelerating queries on free form text from user reviews. The data is currently stored in Parquet with the below schema:  
item\_id INT, user\_id INT, review\_id INT, rating FLOAT, review STRING

The review column contains the full text of the review left by the user. Specifically, the data science team is looking to identify if any of 30 key words exist in this field.

A junior data engineer suggests converting this data to Delta Lake will improve query performance.

Which response to the junior data engineer's suggestion is correct?

- |                                                                                             |                                                                                |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| a) The Delta log creates a term matrix for free text fields to support selective filtering. | b) ZORDER ON review will need to be run to see performance gains.              |
| c) Text data cannot be stored with Delta Lake.                                              | d) Delta Lake statistics are only collected on the first 4 columns in a table. |
| e) Delta Lake statistics are not optimized for free text fields with high cardinality.      |                                                                                |

57. Assuming that the Databricks CLI has been installed and configured correctly, which Databricks CLI command can be used to upload a custom Python Wheel to object storage mounted with the DBFS for use with a production job?

- a) configure
- b) fs
- c) libraries
- d) workspace

58.

The following table consists of items found in user carts within an e-commerce website.

```
Carts (id LONG, items ARRAY<STRUCT<id: LONG, count: INT>>)
id items email
1001[[{"id": "DESK65", count: 1}] "u1@domain.com"
1002[[{"id": "KYBD45", count: 1}, {"id": "M27", count: 2}] "u2@domain.com"
1003[[{"id": "M27", count: 1}] "u3@domain.com"
```

The following MERGE statement is used to update this table using an updates view, with schema evolution enabled on this table.

```
MERGE INTO carts c
USING updates u
ON c.id = u.id
WHEN MATCHED
THEN UPDATE SET *
```

How would the following update be handled?

```
(new nested field, missing existing column)
id items
1001[[{"id": "DESK65", count: 2, coupon: "BOGO50"}]]
```

The following table consists of items found in user carts within an e-commerce website.

(See image on the left)

The following MERGE statement is used to update this table using an updates view, with schema evolution enabled on this table.

(See image on the left)

How would the following update be handled?

(See image on the left)

- a) The new nested field is added to the target schema, and files underlying existing records are updated to include NULL values for the new field.
- b) The update throws an error because changes to existing columns in the target schema are not supported.
- c) The update is moved to a separate "rescued" column because it is missing a column expected in the target schema.
- d) The new nested Field is added to the target schema, and dynamically read as NULL for existing unmatched records.

59. An upstream system is emitting change data capture (CDC) logs that are being written to a cloud object storage directory. Each record in the log indicates the change type (insert, update, or delete) and the values for each field after the change. The source table has a primary key identified by the field `pk_id`.

For auditing purposes, the data governance team wishes to maintain a full record of all values that have ever been valid in the source system. For analytical purposes, only the most recent value for each record needs to be recorded. The Databricks job to ingest these records occurs once per hour, but each individual record may have changed multiple times over the course of an hour.

Which solution meets these requirements?

- a) Iterate through an ordered set of changes to the table, applying each in turn to create the current state of the table (insert, update, delete), timestamp of change, and the values.
- b) Deduplicate records in each batch by `pk_id` and overwrite the target table.
- c) Ingest all log information into a bronze table; use `MERGE INTO` to insert, update, or delete the most recent entry for each `pk_id` into a silver table to recreate the current table state.
- d) Use `MERGE INTO` to insert, update, or delete the most recent entry for each `pk_id` into a bronze table, then propagate all changes throughout the system.
- e) Use Delta Lake's change data feed to automatically process CDC data from an external system, propagating all changes to all dependent tables in the Lakehouse.

60.

The business intelligence team has a dashboard configured to track various summary metrics for retail stores. This includes total sales for the previous day alongside totals and averages for a variety of time periods. The fields required to populate this dashboard have the following schema:

```
store_id INT, total_sales_gtd FLOAT, avg_daily_sales_gtd FLOAT, total_sales_ytd
FLOAT, avg_daily_sales_ytd FLOAT, previous_day_sales FLOAT, total_sales_7d FLOAT,
avg_daily_sales_7d FLOAT, updated TIMESTAMP
```

For demand forecasting, the Lakehouse contains a validated table of all itemized sales updated incrementally in near real-time. This table, named `products_per_order`, includes the following fields:

```
store_id INT, order_id INT, product_id INT, quantity INT, price FLOAT,
order_timestamp TIMESTAMP
```

Because reporting on long-term sales trends is less volatile, analysts using the new dashboard only require data to be refreshed once daily. Because the dashboard will be queried interactively by many users throughout a normal business day, it should return results quickly and reduce total compute associated with each materialization.

Which solution meets the expectations of the end users while controlling and limiting possible costs?

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(See image on the left)

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Because reporting on long-term sales trends is less volatile, analysts using the new dashboard only require data to be refreshed once daily. Because the dashboard will be queried interactively by many users throughout a normal business day, it should return results quickly and reduce total compute associated with each materialization.

Which solution meets the expectations of the end users while controlling and limiting possible costs?

- |                                                                                                                                        |                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| a) Populate the dashboard by configuring a nightly batch job to save the required values as a table overwritten with each update.      | b) Use Structured Streaming to configure a live dashboard against the <code>products_per_order</code> table within a Databricks notebook. |
| c) Use the Delta Cache to persist the <code>products_per_order</code> table in memory to quickly update the dashboard with each query. | d) Define a view against the <code>products_per_order</code> table and define the dashboard against this view.                            |

61. A view is registered with the following code:  
(See image on the left)  
Both users and orders are Delta Lake tables.

Which statement describes the results of querying recent\_orders?

- |                                                                                                                                                  |                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) All logic will execute at query time and return the result of joining the valid versions of the source tables at the time the query began.    | b) All logic will execute when the view is defined and store the result of joining tables to the DBFS; this stored data will be returned when the view is queried. |
| c) All logic will execute at query time and return the result of joining the valid versions of the source tables at the time the query finishes. | d) Results will be computed and cached when the view is defined; these cached results will incrementally update as new records are inserted into source tables.    |

62. A data ingestion task requires a one-TB JSON dataset to be written out to Parquet with a target part-file size of 512 MB. Because Parquet is being used instead of Delta Lake, built-in file-sizing features such as Auto-Optimize & Auto-Compaction cannot be used.

Which strategy will yield the best performance without shuffling data?

- |                                                                                                                                     |                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a) Set spark.sql.files.maxPartitionBytes to 512 MB, ingest the data, execute the narrow transformations, and then write to parquet. | b) Set spark.sql.shuffle.partitions to 2,048 partitions ( $1\text{TB} \times 1024 \times 1024 / 512$ ), ingest the data, execute the narrow transformations, optimize the data by sorting it (which automatically repartitions the data), and then write to parquet. |
| c) Set spark.sql.shuffle.partitions to 512, ingest the data, execute the narrow transformations, and then write to parquet.         | d) Ingest the data, execute the narrow transformations, repartition to 2,048 partitions ( $1\text{TB} \times 1024 \times 1024 / 512$ ), and then write to parquet.                                                                                                   |
- e) Set spark.sql.adaptive.advisoryPartitionSizeInBytes to 512 MB bytes, ingest the data, execute the narrow transformations, coalesce to 2,048 partitions ( $1\text{TB} \times 1024 \times 1024 / 512$ ), and then write to parquet.



63. Which statement regarding stream-static joins and static Delta tables is correct?
- a) Each microbatch of a stream-static join will use the most recent version of the static Delta table as of the job's initialization.
  - b) Stream-static joins cannot use static Delta tables because of consistency issues.
  - c) The checkpoint directory will be used to track state information for the unique keys present in the join.
  - d) The checkpoint directory will be used to track updates to the static Delta table.
64. A junior data engineer has been asked to develop a streaming data pipeline with a grouped aggregation using DataFrame df. The pipeline needs to calculate the average humidity and average temperature for each non-overlapping five-minute interval. Incremental state information should be maintained for 10 minutes for late-arriving data.

Streaming DataFrame df has the following schema:

"device\_id INT, event\_time TIMESTAMP, temp FLOAT, humidity FLOAT"

Code block:

(see image on the left)

Choose the response that correctly fills in the blank within the code block to complete this task.

- a) "event\_time"
- b) window("event\_time"), "5 minutes").alias("time")
- c) lag("event\_time"), "10 minutes").alias("time")
- d) to\_interval("event\_time"), "5 minutes").alias("time")

65. A Structured Streaming job deployed to production has been resulting in higher than expected cloud storage costs. At present, during normal execution, each microbatch of data is processed in less than 3s; at least 12 times per minute, a microbatch is processed that contains 0 records. The streaming write was configured using the default trigger settings. The production job is currently scheduled alongside many other Databricks jobs in a workspace with instance pools provisioned to reduce start-up time for jobs with batch execution.

Holding all other variables constant and assuming records need to be processed in less than 10 minutes, which adjustment will meet the requirement?

- a) Set the trigger interval to 3 seconds; the default trigger interval is consuming too many records per batch, resulting in spill to disk that can increase volume costs.
- b) Set the trigger interval to 500 milliseconds; setting a small but non-zero trigger interval ensures that the source is not queried too frequently.
- c) Set the trigger interval to 10 minutes; each batch calls APIs in the source storage account, so decreasing trigger frequency to maximum allowable threshold should minimize this cost.
- d) Increase the number of shuffle partitions to maximize parallelism, since the trigger interval cannot be modified without modifying the checkpoint directory.
- e) Use the trigger once option and configure a Databricks job to execute the query every 10 minutes; this approach minimizes costs for both compute and storage.

66. Which statement describes Delta Lake optimized writes?

- a) An asynchronous job runs after the write completes to detect if files could be further compacted; if yes, an OPTIMIZE job is executed toward a default of 1 GB.
- b) Optimized writes use logical partitions instead of directory partitions; because partition boundaries are only represented in metadata, fewer small files are written.
- c) Before a Jobs cluster terminates, OPTIMIZE is executed on all tables modified during the most recent job.
- d) Data is queued in a messaging bus instead of committing data directly to memory; all data is committed from the messaging bus in one batch once the job is complete.
- e) A shuffle occurs prior to writing to try to group similar data together resulting in fewer files instead of each executor writing multiple files based on directory partitions.

67. 

```
(spark.readStream
  .format("parquet")
  .load("/mnt/raw_orders/")
  .withWatermark("time", "2 hours")
  .dropDuplicates(["customer_id", "order_id"])
  .writeStream
  .trigger(once=True)
  .table("orders")
)
```

A task orchestrator has been configured to run two hourly tasks. First, an outside system writes Parquet data to a directory mounted at /mnt/raw\_orders/. After this data is written, a Databricks job containing the following code is executed:

(See image on the left)

Assume that the fields customer\_id and order\_id serve as a composite key to uniquely identify each order, and that the time field indicates when the record was queued in the source system.

If the upstream system is known to occasionally enqueue duplicate entries for a single order hours apart, which statement is correct?

- a) The orders table will not contain duplicates, but records arriving more than 2 hours late will be ignored and missing from the table.
- b) Duplicate records arriving more than 2 hours apart will be dropped, but duplicates that arrive in the same batch may both be written to the orders table.
- c) The orders table will contain only the most recent 2 hours of records and no duplicates will be present.
- d) All records will be held in the state store for 2 hours before being deduplicated and committed to the orders table.
- e) Duplicate records enqueued more than 2 hours apart may be retained and the orders table may contain duplicate records with the same customer\_id and order\_id.

68. A data engineer is configuring a pipeline that will potentially see late-arriving, duplicate records.

In addition to de-duplicating records within the batch, which of the following approaches allows the data engineer to deduplicate data against previously processed records as it is inserted into a Delta table?

- a) Set the configuration delta.deduplicate = true.
- b) Perform a full outer join on a unique key and overwrite existing data.
- c) Perform an insert-only merge with a matching condition on a unique key.
- d) VACUUM the Delta table after each batch completes.
- e) Rely on Delta Lake schema enforcement to prevent duplicate records.

69. A DLT pipeline includes the following streaming tables:

- raw\_iot ingests raw device measurement data from a heart rate tracking device.
- bpm\_stats incrementally computes user statistics based on BPM measurements from raw\_iot.

How can the data engineer configure this pipeline to be able to retain manually deleted or updated records in the raw\_iot table, while recomputing the downstream table bpm\_stats table when a pipeline update is run?

- a) Set the skipChangeCommits flag to true on bpm\_stats      b) Set the pipelines.reset.allowed property to false on bpm\_stats
- c) Set the skipChangeCommits flag to true on raw\_iot      d) Set the pipelines.reset.allowed property to false on raw\_iot

70. A data pipeline uses Structured Streaming to ingest data from Apache Kafka to Delta Lake. Data is being stored in a bronze table, and includes the Kafka-generated timestamp, key, and value. Three months after the pipeline is deployed, the data engineering team has noticed some latency issues during certain times of the day.

A senior data engineer updates the Delta Table's schema and ingestion logic to include the current timestamp (as recorded by Apache Spark) as well as the Kafka topic and partition. The team plans to use these additional metadata fields to diagnose the transient processing delays.

Which limitation will the team face while diagnosing this problem?

- a) New fields cannot be added to a production Delta table.      b) Updating the table schema requires a default value provided for each field added.
- c) Spark cannot capture the topic and partition fields from a Kafka source.      d) Updating the table schema will invalidate the Delta transaction log metadata.
- e) New fields will not be computed for historic records.

71. The marketing team is looking to share data in an aggregate table with the sales\_organization, but the field names used by the teams do not match, and a number of marketing-specific fields have not been approved for the sales org.

Which of the following solutions addresses the situation while emphasizing simplicity?

- a) Create a new table with the required schema and use Delta Lake's DEEP CLONE functionality to sync up changes committed to one table to the corresponding table.
- b) Create a view on the marketing table selecting only those fields approved for the sales team; alias the names of any fields that should be standardized to the sales naming conventions.
- c) Use a CTAS statement to create a derivative table from the marketing table; configure a production job to propagate changes.
- d) Add a parallel table to the current production pipeline, updating a new sales table that varies as required from the marketing table.

72. A small company based on the United States has recently contracted a consulting firm in India to implement several new data engineering pipelines to power artificial intelligence applications. All the company's data is stored in regional cloud storage in the United States. The Workspace administrator at the company is uncertain about where the Databricks workspace used by the contractors should be deployed.

Assuming that all data governance considerations are accounted for, which statement accurately informs this decision?

- a) Cross-region reads and writes can incur significant costs and latency; whenever possible, compute should be deployed in the same region the data is stored.
- b) Databricks workspaces do not rely on any regional infrastructure; as such, the decision should be made based upon what is most convenient for the workspace administrator.
- c) Databricks notebooks send all executable code from the user's browser to virtual machines over the open internet; whenever possible, choosing a workspace region near the end users is the most secure.
- d) Databricks runs HDFS on cloud volume storage; as such, cloud virtual machines must be deployed in the region where the data is stored.

73. A CHECK constraint has been successfully added to the Delta table named activity\_details using the following logic:  
(See image on the left)  
A batch job is attempting to insert new records to the table, including a record where latitude = 45.50 and longitude = 212.67.

Which statement describes the outcome of this batch insert?

- a) The write will fail when the violating record is reached; any records previously processed will be recorded to the target table.
- b) The write will insert all records except those that violate the table constraints; the violating records will be recorded to a quarantine table.
- c) The write will fail completely because of the constraint violation and no records will be inserted into the target table.
- d) The write will include all records in the target table; any violations will be indicated in the boolean column named valid\_coordinates.
- e) The write will insert all records except those that violate the table constraints; the violating records will be reported in a warning log.
74. A junior data engineer is migrating a workload from a relational database system to the Databricks Lakehouse. The source system uses a star schema, leveraging foreign key constraints and multi-table inserts to validate records on write.

Which consideration will impact the decisions made by the engineer while migrating this workload?

- a) Databricks only allows foreign key constraints on hashed identifiers, which avoid collisions in highly-parallel writes.
- b) Databricks supports Spark SQL and JDBC; all logic can be directly migrated from the source system without refactoring.
- c) Committing to multiple tables simultaneously requires taking out multiple table locks and can lead to a state of deadlock.
- d) Foreign keys must reference a primary key field; multi-table inserts must leverage Delta Lake's upsert functionality.
- e) All Delta Lake transactions are ACID compliant against a single table, and Databricks does not enforce foreign key constraints.

75. A data architect has heard about Delta Lake's built-in versioning and time travel capabilities. For auditing purposes, they have a requirement to maintain a full record of all valid street addresses as they appear in the customers table.

The architect is interested in implementing a Type 1 table, overwriting existing records with new values and relying on Delta Lake time travel to support long-term auditing. A data engineer on the project feels that a Type 2 table will provide better performance and scalability.

Which piece of information is critical to this decision?

- a) Shallow clones can be combined with Type 1 tables to accelerate historic queries for long-term versioning.
- b) Delta Lake time travel does not scale well in cost or latency to provide a long-term versioning solution.
- c) Delta Lake time travel cannot be used to query previous versions of these tables because Type 1 changes modify data files in place.
- d) Delta Lake only supports Type 0 tables; once records are inserted to a Delta Lake table, they cannot be modified.
- e) Data corruption can occur if a query fails in a partially completed state because Type 2 tables require setting multiple fields in a single update.

76.

```
.groupBy(
  window("event_time", "5 minutes"),
  "id")
  .count()
).withWatermark("event_time", 2 hours)
impressions.join(clicks, expr("clickAdId = impressionAdId"), "inner")
```

A data engineer wants to join a stream of advertisement impressions (when an ad was shown) with another stream of user clicks on advertisements to correlate when impressions led to monetizable clicks. In the code below, Impressions is a streaming DataFrame with a watermark ("event\_time", "10 minutes")

(See image on the left)

The data engineer notices the query slowing down significantly.

Which solution would improve the performance?

- a) Joining on event time constraint:  $\text{clickTime} \geq \text{impressionTime} - \text{interval } 3 \text{ hours}$  and removing watermarks
- b) Joining on event time constraint:  $\text{clickTime} + 3 \text{ hours} < \text{impressionTime} - 2 \text{ hours}$
- c) Joining on event time constraint:  $\text{clickTime} \geq \text{impressionTime}$  AND  $\text{clickTime} \leq \text{impressionTime} + \text{interval } 1 \text{ hour}$
- d) Joining on event time constraint:  $\text{clickTime} == \text{impressionTime}$  using a leftOuter join

77. A junior data engineer has manually configured a series of jobs using the Databricks Jobs UI. Upon reviewing their work, the engineer realizes that they are listed as the "Owner" for each job. They attempt to transfer "Owner" privileges to the "DevOps" group, but cannot successfully accomplish this task.

Which statement explains what is preventing this privilege transfer?

- a) A user can only transfer job ownership to a group if they are also a member of that group.
- b) Other than the default "admins" group, only individual users can be granted privileges on jobs.
- c) Databricks jobs must have exactly one owner; "Owner" privileges cannot be assigned to a group.
- d) The creator of a Databricks job will always have "Owner" privileges; this configuration cannot be changed.
- e) Only workspace administrators can grant "Owner" privileges to a group.

78. 

```
CREATE VIEW user_ltv_no_minors AS
SELECT email, age, ltv
FROM user_ltv
WHERE
  CASE
    WHEN is_member("auditing") THEN TRUE
    ELSE age >= 18
  END
```

A table named user\_ltv is being used to create a view that will be used by data analysts on various teams. Users in the workspace are configured into groups, which are used for setting up data access using ACLs.

The user\_ltv table has the following schema:

email STRING, age INT, ltv INT

The following view definition is executed:

(See image on the left)

An analyst who is not a member of the marketing group executes the following query:

```
SELECT * FROM user_ltv_no_minors
```

Which statement describes the results returned by this query?

- a) All values for the age column will be returned as null values, all other columns will be returned with the values in user\_ltv.
- b) All age values less than 18 will be returned as null values, all other columns will be returned with the values in user\_ltv.
- c) All columns will be displayed normally for those records that have an age greater than 18; records not meeting this condition will be omitted.
- d) All columns will be displayed normally for those records that have an age greater than 17; records not meeting this condition will be omitted.



79. All records from an Apache Kafka producer are being ingested into a single Delta Lake table with the following schema:

key BINARY, value BINARY, topic STRING, partition LONG, offset LONG, timestamp LONG

There are 5 unique topics being ingested. Only the "registration" topic contains Personal Identifiable Information (PII). The company wishes to restrict access to PII. The company also wishes to only retain records containing PII in this table for 14 days after initial ingestion. However, for non-PII information, it would like to retain these records indefinitely.

Which of the following solutions meets the requirements?

- a) All data should be deleted biweekly; Delta Lake's time travel functionality should be leveraged to maintain a history of non-PII information.
- b) Because the value field is stored as binary data, this information is not considered PII and no special precautions should be taken.
- c) Data should be partitioned by the topic field, allowing ACLs and delete statements to leverage partition boundaries.
- d) Data should be partitioned by the registration field, allowing ACLs and delete statements to be set for the PII directory.
- e) Separate object storage containers should be specified based on the partition field, allowing isolation at the storage level.

80.

```
(spark.read
  .format("delta")
  .option("readChangeData", True)
  .option("startingTimestamp", '2021-08-22 00:00:00')
  .option("endingTimestamp", '2021-08-29 00:00:00')
  .table("user_lookup")
  .createOrReplaceTempView("changes"))

spark.sql("""
DELETE FROM user_aggregates
WHERE user_id IN (
  SELECT user_id
  FROM changes
  WHERE _change_type='delete'
)
""")
```

The data governance team is reviewing code used for deleting records for compliance with GDPR. The following logic has been implemented to propagate delete requests from the user\_lookup table to the user\_aggregates table.

(see image on the left)

Assuming that user\_id is a unique identifying key and that all users that have requested deletion have been removed from the user\_lookup table, which statement describes whether successfully executing the above logic guarantees that the records to be deleted from the user\_aggregates table are no longer accessible and why?

- a) No; the change data feed only tracks inserts and updates, not deleted records.
- b) Yes; Delta Lake ACID guarantees provide assurance that the DELETE command succeeded fully and permanently purged these records.
- c) No; files containing deleted records may still be accessible with time travel until a VACUUM command is used to remove invalidated data files.
- d) No; the Delta Lake DELETE command only provides ACID guarantees when combined with the MERGE INTO command.
- e) Yes; the change data feed uses foreign keys to ensure delete consistency throughout the Lakehouse.

81. An external object storage container has been mounted to the location  
/mnt/finance\_eda\_bucket.

The following logic was executed to create a database for the finance team:

(See image on the left)

After the database was successfully created and permissions configured, a member of the finance team runs the following code:

(See image on the left)

If all users on the finance team are members of the finance group, which statement describes how the tx\_sales table will be created?

- a) An managed table will be created in the storage container mounted to /mnt/finance\_eda\_bucket.
- b) An external table will be created in the storage container mounted to /mnt/finance\_eda\_bucket.
- c) A managed table will be created in the DBFS root storage container.
- d) A logical table will persist the physical plan to the Hive Metastore in the Databricks control plane.
- e) A logical table will persist the query plan to the Hive Metastore in the Databricks control plane.

82. The data engineering team has been tasked with configuring connections to an external database that does not have a supported native connector with Databricks. The external database already has data security configured by group membership. These groups map directly to user groups already created in Databricks that represent various teams within the company.

A new login credential has been created for each group in the external database. The Databricks Utilities Secrets module will be used to make these credentials available to Databricks users.

Assuming that all the credentials are configured correctly on the external database and group membership is properly configured on Databricks, which statement describes how teams can be granted the minimum necessary access to using these credentials?

- a) "Read" permissions should be set on a secret scope containing only those credentials that will be used by a given team.
- b) No additional configuration is necessary as long as all users are configured as administrators in the workspace where secrets have been added.
- c) "Read" permissions should be set on a secret key mapped to those credentials that will be used by a given team.
- d) "Manage" permissions should be set on a secret scope containing only those credentials that will be used by a given team.
- e) "Manage" permissions should be set on a secret key mapped to those credentials that will be used by a given team.

83. What statement is true regarding the retention of job run history?

- a) It is retained until you export or delete job run logs
- b) It is retained for 90 days or until the run-id is re-used through custom run configuration
- c) It is retained for 60 days, during which you can export notebook run results to HTML
- d) It is retained for 60 days, after which logs are archived
- e) It is retained for 30 days, during which time you can deliver job run logs to DBFS or S3

84. A data engineer, User A, has promoted a new pipeline to production by using the REST API to programmatically create several jobs. A DevOps engineer, User B, has configured an external orchestration tool to trigger job runs through the REST API. Both users authorized the REST API calls using their personal access tokens.

Which statement describes the contents of the workspace audit logs concerning these events?

- a) Because User A created the jobs, their identity will be associated with both the job creation events and the job run events.
  - b) Because the REST API was used for job creation and triggering runs, a Service Principal will be automatically used to identify these events.
  - c) Because these events are managed separately, User A will have their identity associated with the job creation events and User B will have their identity associated with the job run events.
  - d) Because the REST API was used for job creation and triggering runs, user identity will not be captured in the audit logs.
85. A distributed team of data analysts share computing resources on an interactive cluster with autoscaling configured. In order to better manage costs and query throughput, the workspace administrator is hoping to evaluate whether cluster upscaling is caused by many concurrent users or resource-intensive queries.

In which location can one review the timeline for cluster resizing events?

- a) Workspace audit logs
  - b) Ganglia
  - c) Executor's log file
  - d) Cluster Event Log
  - e) Driver's log file
86. When evaluating the Ganglia Metrics for a given cluster with 3 executor nodes, which indicator would signal proper utilization of the VM's resources?
- a) The five Minute Load Average remains consistent/flat
  - b) Bytes Received never exceeds 80 million bytes per second
  - c) Network I/O never spikes
  - d) Total Disk Space remains constant
  - e) CPU Utilization is around 75%

87. The data engineer is using Spark's MEMORY\_ONLY storage level.

Which indicators should the data engineer look for in the Spark UI's Storage tab to signal that a cached table is not performing optimally?

- a) Size on Disk is > 0
- b) The RDD Block Name includes the "\*" annotation signaling a failure to cache
- c) The number of Cached Partitions > the number of Spark Partitions
- d) On Heap Memory Usage is within 75% of Off Heap Memory Usage

88. Review the following error traceback:

(See image on the left)

Which statement describes the error being raised?

- a) There is a type error because a DataFrame object cannot be multiplied.
- b) There is a syntax error because the heartrate column is not correctly identified as a column.
- c) There is no column in the table named heartrateheartrateheartrate
- d) The code executed was PySpark but was executed in a Scala notebook.
- e) There is a type error because a column object cannot be multiplied.

89. Which describes a method of installing a Python package scoped at the notebook level to all nodes in the currently active cluster?

- a) Use `b` in a notebook cell
- b) Use `%sh pip install` in a notebook cell
- c) Install libraries from PyPI using the cluster UI
- d) Run `source env/bin/activate` in a notebook setup script
- e) Use `%pip install` in a notebook cell

90. What is the first line of a Databricks Python notebook when viewed in a text editor?

- a) `%python`
- b) `// Databricks notebook source`
- c) `# Databricks notebook source`
- d) `-- Databricks notebook source`
- e) `# MAGIC %python`

91. Incorporating unit tests into a PySpark application requires upfront attention to the design of your jobs, or a potentially significant refactoring of existing code.  
Which statement describes a main benefit that offset this additional effort?
- a) Yields faster deployment and execution times
  - b) Improves the quality of your data
  - c) Troubleshooting is easier since all steps are isolated and tested individually
  - d) Validates a complete use case of your application
  - e) Ensures that all steps interact correctly to achieve the desired end result
92. What describes integration testing?
- a) It validates an application use case.
  - b) It validates interactions between subsystems of your application.
  - c) It validates behavior of individual elements of an application.
  - d) It requires an automated testing framework.
93. The Databricks CLI is used to trigger a run of an existing job by passing the job\_id parameter. The response that the job run request has been submitted successfully includes a field run\_id. Which statement describes what the number alongside this field represents?
- a) The globally unique ID of the newly triggered run.
  - b) The total number of jobs that have been run in the workspace.
  - c) The job\_id is returned in this field.
  - d) The number of times the job definition has been run in this workspace.
  - e) The job\_id and number of times the job has been run are concatenated and returned.

94. A Databricks job has been configured with 3 tasks, each of which is a Databricks notebook. Task A does not depend on other tasks. Tasks B and C run in parallel, with each having a serial dependency on Task A.

If task A fails during a scheduled run, which statement describes the results of this run?

- a) Unless all tasks complete successfully, no changes will be committed to the Lakehouse; because task A failed, all commits will be rolled back automatically.
- b) Tasks B and C will be skipped; some logic expressed in task A may have been committed before task failure.
- c) Because all tasks are managed as a dependency graph, no changes will be committed to the Lakehouse until all tasks have successfully been completed.
- d) Tasks B and C will be skipped; task A will not commit any changes because of stage failure.
- e) Tasks B and C will attempt to run as configured; any changes made in task A will be rolled back due to task failure.

95. A Databricks job has been configured with 3 tasks, each of which is a Databricks notebook. Task A does not depend on other tasks. Tasks B and C run in parallel, with each having a serial dependency on Task A.

What will be the resulting state if tasks A and B complete successfully, but task C fails during a scheduled run?

- a) All logic expressed in the notebook associated with task A and B will have been successfully completed; some operations in task C may have completed successfully.
- b) All logic expressed in the notebook associated with Task A and B will have been successfully completed; any changes made in task C will be rolled back due to task failure.
- c) Unless all tasks complete successfully, no changes will be committed to the Lakehouse; because task C failed, all commits will be rolled back automatically.
- d) Because all tasks are managed as a dependency graph, no changes will be committed to the Lakehouse until all tasks have successfully been completed.



96. You are testing a collection of mathematical functions, one of which calculates the area under a curve as described by another function.

```
assert(myIntegrate(lambda x: x*x, 0, 3) [0] == 9)
```

Which kind of test would the above line exemplify?

- a) End-to-end
  - b) Integration
  - c) Manual
  - d) Functional
  - e) Unit
97. In order to prevent accidental commits to production data, a senior data engineer has instituted a policy that all development work will reference clones of Delta Lake tables. After testing both DEEP and SHALLOW CLONE, development tables are created using SHALLOW CLONE.

A few weeks after initial table creation, the cloned versions of several tables implemented as Type 1 Slowly Changing Dimension (SCD) stop working. The transaction logs for the source tables show that VACUUM was run the day before.

Which statement describes why the cloned tables are no longer working?

- a) Tables created with SHALLOW CLONE are automatically deleted after their default retention threshold of 7 days.
- b) The data files compacted by VACUUM are not tracked by the cloned metadata; running REFRESH on the cloned table will pull in recent changes.
- c) Running VACUUM automatically invalidates any shallow clones of a table; DEEP CLONE should always be used when a cloned table will be repeatedly queried.
- d) The metadata created by the CLONE operation is referencing data files that were purged as invalid by the VACUUM command.
- e) Because Type 1 changes overwrite existing records, Delta Lake cannot guarantee data consistency for cloned tables.

98. You are performing a join operation to combine values from a static userLookup table with a streaming DataFrame streamingDF.

Which code block attempts to perform an invalid stream-static join?

- a) streamingDF.join(userLookup, ["user\_id"], how="outer")
  - b) userLookup.join(streamingDF, ["userid"], how="inner")
  - c) userLookup.join(streamingDF, ["user\_id"], how="right")
  - d) streamingDF.join(userLookup, ["userid"], how="inner")
  - e) streamingDF.join(userLookup, ["user\_id"], how="left")
99. Spill occurs as a result of executing various wide transformations. However, diagnosing spill requires one to proactively look for key indicators.

Where in the Spark UI are two of the primary indicators that a partition is spilling to disk?

- a) Executor's detail screen and Executor's log files
  - b) Stage's detail screen and Executor's log files
  - c) Stage's detail screen and Query's detail screen
  - d) Query's detail screen and Job's detail screen
  - e) Driver's and Executor's log files
100. Which indicators would you look for in the Spark UI's Storage tab to signal that a cached table is not performing optimally? Assume you are using Spark's MEMORY\_ONLY storage level.
- a) Size on Disk is < Size in Memory
  - b) On Heap Memory Usage is within 75% of Off Heap Memory Usage
  - c) The number of Cached Partitions > the number of Spark Partitions
  - d) The RDD Block Name includes the "\*" annotation signaling a failure to cache
  - e) Size on Disk is > 0

101. Which statement describes a key benefit of an end-to-end test?

- a) Pinpoints errors in the building blocks of your application
- b) Makes it easier to automate your test suite
- c) Closely simulates real world usage of your application
- d) Ensures code is optimized for a real-life workflow
- e) Provides testing coverage for all code paths and branches

102. A nightly batch job is configured to ingest all data files from a cloud object storage container where records are stored in a nested directory structure YYYY/MM/DD. The data for each date represents all records that were processed by the source system on that date, noting that some records may be delayed as they await moderator approval. Each entry represents a user review of a product and has the following schema:

user\_id STRING, review\_id BIGINT, product\_id BIGINT, review\_timestamp TIMESTAMP, review\_text STRING

The ingestion job is configured to append all data for the previous date to a target table reviews\_raw with an identical schema to the source system. The next step in the pipeline is a batch write to propagate all new records inserted into reviews\_raw to a table where data is fully deduplicated, validated, and enriched.

Which solution minimizes the compute costs to propagate this batch of data?

- a) Perform a batch read on the reviews\_raw table and perform an insert-only merge using the natural composite key user\_id, review\_id, product\_id, review\_timestamp.
- b) Reprocess all records in reviews\_raw and overwrite the next table in the pipeline.
- c) Use Delta Lake version history to get the difference between the latest version of reviews\_raw and one version prior, then write these records to the next table.
- d) Configure a Structured Streaming read against the reviews\_raw table using the trigger once execution mode to process new records as a batch job.
- e) Filter all records in the reviews\_raw table based on the review\_timestamp; batch append those records produced in the last 48 hours.

103. Which statement describes the default execution mode for Databricks Auto Loader?

- a) New files are identified by listing the input directory; new files are incrementally and idempotently loaded into the target Delta Lake table.
- b) New files are identified by listing the input directory; the target table is materialized by directly querying all valid files in the source directory.
- c) Webhooks trigger a Databricks job to run anytime new data arrives in a source directory; new data are automatically merged into target tables using rules inferred from the data.
- d) Cloud vendor-specific queue storage and notification services are configured to track newly arriving files; the target table is materialized by directly querying all valid files in the source directory.
- e) Cloud vendor-specific queue storage and notification services are configured to track newly arriving files; new files are incrementally and idempotently loaded into the target Delta Lake table.

104. A Delta Lake table representing metadata about content posts from users has the following schema: user\_id LONG, post\_text STRING, post\_id STRING, longitude FLOAT, latitude FLOAT, post\_time TIMESTAMP, date DATE

Based on the above schema, which column is a good candidate for partitioning the Delta Table?

- a) date
- b) user\_id
- c) post\_id
- d) latitude
- e) post\_time

105. A large company seeks to implement a near real-time solution involving hundreds of pipelines with parallel updates of many tables with extremely high volume and high velocity data.

Which of the following solutions would you implement to achieve this requirement?

- a) Configure Databricks to save all data to attached SSD volumes instead of object storage, increasing file I/O significantly.
- b) Isolate Delta Lake tables in their own storage containers to avoid API limits imposed by cloud vendors.
- c) Store all tables in a single database to ensure that the Databricks Catalyst Metastore can load balance overall throughput.
- d) Use Databricks High Concurrency clusters, which leverage optimized cloud storage connections to maximize data throughput.
- e) Partition ingestion tables by a small time duration to allow for many data files to be written in parallel.